Bill shape in *Pygoscelis* penguins: a reproducible example

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1. Introduction

Three penguin species of the genus *Pygoscelis* breed on the islands near Palmer Station, on the Antarctic Peninsula. K. B. Gorman, T. D. Williams, and W. R. Fraser [1] measured adults of all three over three seasons. The data are openly available [2] and are used here to show a document whose numbers come straight from the analysis.

2. Methods

The analysis uses the 344 penguins in the dataset (152 Adélie, 68 Chinstrap and 124 Gentoo). Bill depth was regressed on bill length, once for the pooled sample and once within each species.

3. Results

Gentoo penguins were the heaviest, at 5076 g on average, against 3701 g for Adélie and 3733 g for Chinstrap (Figure 1).

Pooling the three species, longer bills appear shallower: the slope is −0.085 mm per mm. Within every species the relationship is the opposite, with slopes between 0.179 and 0.222 (Figure 2). The pooled trend is produced by the differences between species, a case of Simpson's paradox.

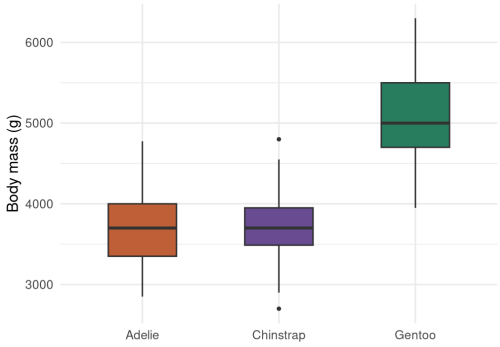


Figure 1: Body mass by species.

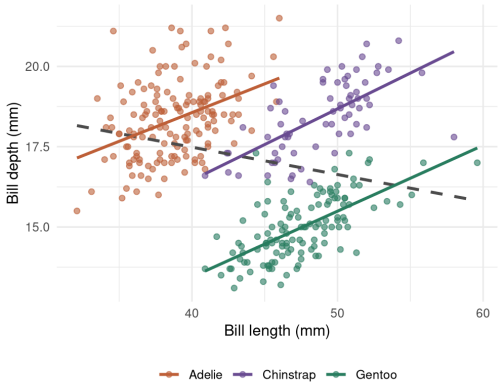


Figure 2: Bill depth against bill length. Dashed line: all species pooled; solid lines: within each species.

4. Discussion

Every number in this text is read from `results/numbers.json`, which the analysis writes. If the data change, the analysis runs again and the paper, the report and the slides update together.

Bibliography

[1] K. B. Gorman, T. D. Williams, and W. R. Fraser, "Ecological Sexual Dimorphism and Environmental Variability within a Community of Antarctic Penguins (Genus Pygoscelis)," *PLoS ONE*, vol. 9, no. 3, p. e90081, 2014, doi: 10.1371/journal.pone.0090081.

[2] A. M. Horst, A. P. Hill, and K. B. Gorman, "palmerpenguins: Palmer Archipelago (Antarctica) penguin data." 2020. doi: 10.5281/zenodo.3960218.